**KAIST AI.61600 DLT: Final Report** **Spring 2026**

# Adapter Optimization Between Frozen Modules: A Controlled NTK Study

*Student name(s): Haechan Kim*

**Abstract**

Modern speech-language systems often connect a frozen speech encoder to a frozen language model through a small trainable adapter. Empirically, the performance of such systems depends strongly on the particular encoder, adapter, and downstream model, but directly studying this interaction with full audio and LLM training is too expensive for a controlled theory project. This report isolates one component: the optimization of an adapter trained between two frozen modules. The full system is replaced by a teacher-student surrogate in which the encoder is a Gaussian representation distribution and the downstream model is a fixed head. The main claim is that adapter width alone does not determine optimization difficulty; the conditioning of the frozen representation and the frozen head enters through the empirical neural tangent kernel (NTK). In the surrogate, the NTK decomposes into terms involving the representation Gram matrix and the frozen-head Gram matrix. Experiments confirm a strong width effect and show that poor encoder or head conditioning decreases the relevant NTK eigenvalues and leaves a larger training error, especially in the moderate-to-wide adapter regime. Additional joint-conditioning, width-threshold, and kernel-drift experiments show that the two frozen modules interact through the spectrum, that ill-conditioned pairs require substantially wider adapters, and that the empirical NTK becomes more stable as width increases.

## 1 Introduction

Pretrained speech-language models can be viewed as a composition

$$x \xrightarrow{E} z \in \mathbb{R}^{d_E} \xrightarrow{A_\theta} u \in \mathbb{R}^{d_L} \xrightarrow{\phi} y \in \mathbb{R}^{d_Y},$$

where $E$ is a frozen audio encoder, $\phi$ is a frozen downstream model, and only the adapter $A_\theta$ is trained. This setting is motivated by empirical evidence that encoder rankings and adapter choices do not transfer cleanly across audio-language systems [Yang et al., 2021, Zhang et al., 2025, Lei et al., 2026]. This report focuses on the optimization aspect of this phenomenon: *when is adapter-only training easy or difficult, given fixed surrounding modules?*

The main simplification is to replace real audio and LLMs by a tractable surrogate that preserves the chain-rule structure of the end-to-end Jacobian. This follows a standard deep-learning-theory workflow: choose a phenomenon, strip it down to a model whose quantities are observable, and analyze that model. The resulting question is narrow but testable: how do the covariance of frozen representations and the Jacobian of the frozen head affect the empirical NTK and gradient-descent behavior of the adapter?

## 2 Problem Setup

The experimental dataset is $\mathcal{D} = \{(z_i, y_i)\}_{i=1}^n$, where $z_i$ is the frozen encoder representation and $y_i$ is a vector-valued label. The encoder is modeled by

$$z_i \sim \mathcal{N}(0, \Sigma_E).$$

Labels are generated by a teacher adapter $A_\star \in \mathbb{R}^{d_L \times d_E}$ and a fixed downstream head $\phi : \mathbb{R}^{d_L} \to \mathbb{R}^{d_Y}$:

$$y_i = \phi(A_\star z_i) + \xi_i, \qquad \xi_i \sim \mathcal{N}(0, \sigma^2 I_{d_Y}).$$

The student adapter is a two-layer ReLU network

$$A_\theta(z) = \frac{1}{\sqrt{m}} W_2 \sigma(W_1 z), \qquad W_1 \in \mathbb{R}^{m \times d_E}, \quad W_2 \in \mathbb{R}^{d_L \times m},$$

where $m$ is the adapter width and $\theta = (W_1, W_2)$ are the only trainable parameters. The predictor is $f_\theta(z) = \phi(A_\theta(z))$, trained by full-batch gradient descent on

$$\widehat{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \|f_\theta(z_i) - y_i\|_2^2.$$

Let $J_0 = \nabla_\theta f_{\theta_0}(\mathcal{D})$ be the Jacobian of the vectorized model outputs at initialization and $K_0 = J_0 J_0^\top$ the empirical NTK. The empirical representation covariance is $\widehat{\Sigma}_z = n^{-1}\sum_i z_i z_i^\top$. The frozen-head sensitivity is summarized by

$$H_\phi = \frac{1}{n}\sum_{i=1}^{n} D\phi(u_i) D\phi(u_i)^\top, \qquad u_i = A_{\theta_0}(z_i),$$

which becomes $H_\phi = CC^\top$ for a linear head $\phi(u) = Cu$. For a positive definite matrix $M$, $\kappa(M) = \lambda_{\max}(M)/\lambda_{\min}(M)$ denotes its condition number.

# 3 NTK Analysis

The optimization hypothesis is that adapter training is governed by the small eigenvalues of $K_0$. The following standard NTK-style statement makes the role of this eigenvalue explicit [Jacot et al., 2018, Du et al., 2019].

**Proposition 3.1 (Linearized NTK convergence).** *Consider the linearized dynamics around initialization, where the residual vector evolves as $r_{t+1} = (I - \eta K_0) r_t$. If $\lambda_0 = \lambda_{\min}(K_0) > 0$ and $\eta \leq 1/\lambda_{\max}(K_0)$, then*

$$\|r_{t+1}\|_2^2 \leq (1 - \eta\lambda_0)^2 \|r_t\|_2^2.$$

**Proof:** Since $K_0$ is symmetric positive semidefinite, the eigenvalues of $I - \eta K_0$ lie in $[0, 1 - \eta\lambda_0]$ under the stated step-size condition. Therefore

$$\|r_{t+1}\|_2 = \|(I - \eta K_0) r_t\|_2 \leq (1 - \eta\lambda_0)\|r_t\|_2,$$

and squaring both sides gives the claim. □

Standard wide-network NTK analyses justify using this linearized dynamics when the empirical kernel does not move much during training [Jacot et al., 2018, Du et al., 2019]. Thus the small eigenvalues of $K_0$ are a natural diagnostic for adapter optimization.

For the linear-head surrogate, the NTK can be written explicitly. Let $c_a^\top$ be row $a$ of $C$, $w_{2,r}$ be column $r$ of $W_2$, $h_i = \sigma(W_1 z_i)$, and $s_{ir} = \mathbf{1}\{(W_1 z_i)_r > 0\}$. For samples $i, j$ and output coordinates $a, b$,

$$\begin{aligned} K_{0,(i,a),(j,b)} &= \frac{1}{m}\langle h_i, h_j\rangle (CC^\top)_{ab} \\ &\quad + \frac{1}{m}\sum_{r=1}^{m} s_{ir} s_{jr}\, \langle z_i, z_j\rangle\, (c_a^\top w_{2,r})(c_b^\top w_{2,r}). \end{aligned}$$

This identity shows why both frozen modules matter. The representation Gram matrix enters through $\langle z_i, z_j\rangle$, while the downstream head enters through $CC^\top$ and the projected adapter columns $Cw_{2,r}$. Thus poor conditioning of either the representation distribution or the frozen head can shrink the useful spectrum of $K_0$ and require a larger width $m$ to obtain comparable convergence. This analysis leads to three empirical predictions. First, increasing $m$ should improve optimization by making the random adapter features more expressive and by stabilizing the empirical kernel. Second, worse conditioning of either $\widehat{\Sigma}_z$ or $H_\phi$ should reduce small NTK eigenvalues and increase the residual error at fixed width. Third, if both frozen modules are ill-conditioned, the error should be worse than in the corresponding one-sided settings because both Gram factors enter the same Jacobian product. The experiments below test these predictions directly.

| Experiment | Setting | $m = 16$ | $m = 128$ | $m = 512$ |
|---|---|---|---|---|
| E1 train MSE | $\kappa_E = 1$ | $2.28{\times}10^{-2}$ | $1.59{\times}10^{-3}$ | $1.36{\times}10^{-4}$ |
| E1 train MSE | $\kappa_E = 100$ | $1.52{\times}10^{-2}$ | $3.03{\times}10^{-3}$ | $5.93{\times}10^{-4}$ |
| E2 train MSE | $\kappa_H = 1$ | $2.40{\times}10^{-2}$ | $1.56{\times}10^{-3}$ | $1.17{\times}10^{-4}$ |
| E2 train MSE | $\kappa_H = 100$ | $1.96{\times}10^{-2}$ | $3.27{\times}10^{-3}$ | $1.78{\times}10^{-3}$ |

**Table 1.** Representative final training MSE values. Width improves optimization in both sweeps, but ill-conditioned frozen modules leave larger residual error at the same width.

# 4 Experiments

All experiments use generated representation-label data rather than raw audio. Unless stated otherwise, the default setting is $d_E = 64$, $d_L = 32$, $d_Y = 8$, $n_{\text{train}} = 512$, $n_{\text{val}} = 512$, $n_{\text{test}} = 4096$, and noise level $\sigma = 0.05$. Each configuration is averaged over five seeds. The reported optimization metric is final training MSE after 800 full-batch gradient steps with the learning rate capped at 1.0.

**E1: encoder conditioning.** The head is fixed with $\kappa(H_\phi) = 1$, while $\Sigma_E$ is diagonal with log-spaced eigenvalues and $\kappa_E \in \{1, 3, 10, 30, 100\}$. The adapter width is $m \in \{16, 32, 64, 128, 256, 512\}$.

**E2: frozen-head conditioning.** The encoder distribution is isotropic, $\Sigma_E = I$, while the linear head $C$ is generated so that $\kappa(CC^\top) \in \{1, 3, 10, 30, 100\}$. The same width grid is used.

**E3: NTK predictiveness.** E1 and E2 are pooled to test whether $\lambda_{\min}(K_0)$ predicts optimization error beyond width. The analysis computes correlations between $\log \lambda_{\min}(K_0)$ and log final training MSE and fits simple log-linear least-squares predictors.

**E4: joint conditioning.** The one-dimensional sweeps are not sufficient to test whether the two frozen modules interact. E4 therefore sweeps $(\kappa_E, \kappa_H) \in \{1, 10, 100\} \times \{1, 10, 100\}$ at widths $m \in \{128, 512\}$ using three seeds. This experiment checks whether the easiest and hardest regimes are organized by the joint spectrum of the representation and head factors rather than by either factor alone.

**E5: width threshold.** To test whether conditioning changes the amount of overparameterization needed, E5 fixes four frozen-module pairs: $(1, 1)$, $(100, 1)$, $(1, 100)$, and $(100, 100)$ for $(\kappa_E, \kappa_H)$. The width is swept over $m \in \{32, 64, 128, 256, 512, 1024\}$ with three seeds. The target error budget is final train MSE $10^{-3}$.

**E6: kernel drift.** The NTK analysis uses the initialization kernel $K_0$, so E6 measures how far the empirical kernel moves during nonlinear training. For the well-conditioned setting $(\kappa_E, \kappa_H) = (1, 1)$, the ill-conditioned-encoder setting $(100, 1)$, and the ill-conditioned-head setting $(1, 100)$, the experiment computes

$$\frac{\|K_T - K_0\|_F}{\|K_0\|_F},$$

where $K_T$ is the empirical NTK at the final iterate. Because this requires two exact NTK computations per run, E6 uses $n_{\text{train}} = 256$, $n_{\text{test}} = 2048$, $n_{\text{NTK}} = 32$, widths $m \in \{64, 128, 256, 512\}$, and three seeds.

# 5 Results

Table 1 and Figure 1 show the two main sweeps. Width is the dominant factor: increasing $m$ from 16 to 512 reduces final training MSE by one to two orders of magnitude. Conditioning matters once the adapter is moderately wide. In E1, at $m = 512$, the final MSE increases from $1.36 \times 10^{-4}$ at $\kappa_E = 1$ to $5.93 \times 10^{-4}$ at $\kappa_E = 100$; the corresponding mean $\lambda_{\min}(K_0)$ decreases from 9.39 to 6.85. In E2 the effect is stronger: at $m = 512$, the final MSE increases from $1.17 \times 10^{-4}$ at $\kappa_H = 1$ to $1.78 \times 10^{-3}$ at $\kappa_H = 100$, while $\lambda_{\min}(K_0)$ decreases from 9.62 to 0.46.

Figure 2 shows that the NTK diagnostic is predictive, but not the only explanatory variable. Across all 300 E1/E2 runs, the Pearson correlation between $\log \lambda_{\min}(K_0)$ and log final train MSE is $-0.759$. Width alone already explains much of the variance in this controlled experiment ($R^2 = 0.876$ for a log-linear width-only predictor), while adding conditioning information increases the fit to $R^2 = 0.916$. The NTK signal becomes strongest in the overparameterized regime: within the fixed width slice $m = 512$, the combined E1/E2 correlation between $\log \lambda_{\min}(K_0)$ and log final train MSE is $-0.896$ (and $-0.969$ in E2 alone). This is consistent with the theoretical picture: after width is large enough to remove severe underparameterization, the frozen-module spectrum becomes a meaningful predictor of residual optimization error.

The joint-conditioning sweep provides a stronger test of the hypothesis because it varies both frozen modules simultaneously. Increasing either $\kappa_E$ or $\kappa_H$ raises the final error and generally reduces $\lambda_{\min}(K_0)$,

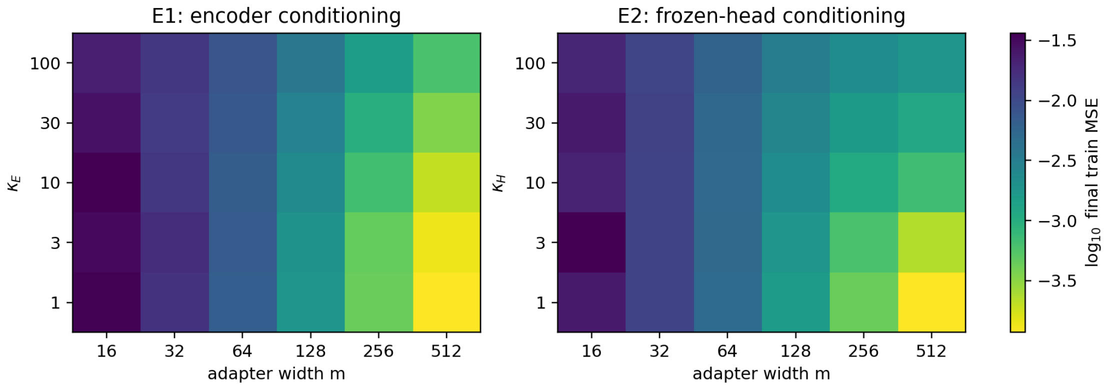


**Figure 1.** Final train MSE over width and conditioning. Darker values correspond to smaller MSE. Both encoder-side and head-side ill-conditioning make optimization harder at fixed width.

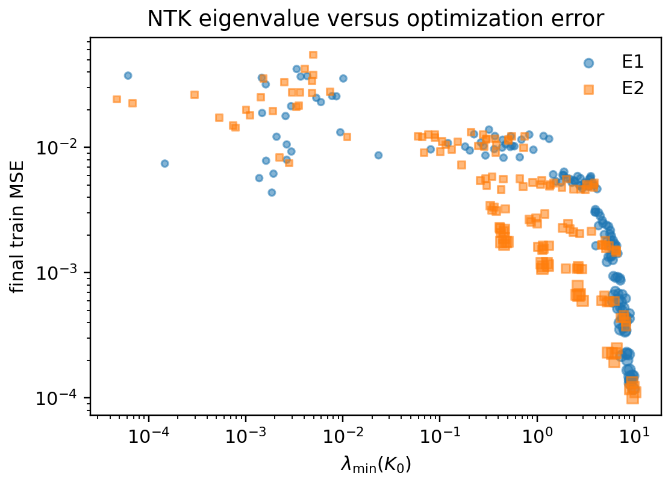


**Figure 2.** Relationship between the empirical NTK minimum eigenvalue and final training error for E1 and E2. Marker size increases with adapter width.

with the largest degradation when both are ill-conditioned. At $m = 512$, the well-conditioned setting has final train MSE $1.14 \times 10^{-4}$ and mean $\lambda_{\min}(K_0) = 9.45$. With only encoder ill-conditioning $(\kappa_E, \kappa_H) = (100, 1)$, the MSE rises to $5.30 \times 10^{-4}$ and $\lambda_{\min}(K_0)$ drops to 6.64. With only head ill-conditioning $(1, 100)$, the MSE rises further to $1.77 \times 10^{-3}$ and $\lambda_{\min}(K_0)$ drops to 0.465. When both modules are ill-conditioned $(100, 100)$, the MSE is $2.34 \times 10^{-3}$ and $\lambda_{\min}(K_0)$ is only 0.306. Across the E4 runs, the correlation between $\log \lambda_{\min}(K_0)$ and log final train MSE is $-0.793$ at $m = 128$ and $-0.915$ at $m = 512$, again showing that the NTK spectrum becomes most informative in the wider regime.

The width-threshold experiment makes this conclusion operational. With target train MSE $10^{-3}$, the well-conditioned pair reaches the target at $m = 256$, while the encoder-bad pair requires $m = 512$. The head-bad pair and the both-bad pair do not reach the target even at $m = 1024$; their best observed MSE values are $1.55 \times 10^{-3}$ and $1.99 \times 10^{-3}$, respectively. The corresponding $\lambda_{\min}(K_0)$ values at $m = 1024$ are 10.38 for the well-conditioned pair, 7.45 for the encoder-bad pair, 0.497 for the head-bad pair, and 0.339 for the both-bad pair. Thus adding width helps every condition, but poor head-side or joint conditioning leaves the effective kernel spectrum too small to match the well-conditioned optimization behavior within the tested width budget.

E6 checks that the preceding NTK interpretation is most appropriate at larger width. The relative kernel drift decreases nearly monotonically with $m$ in all three tested conditions. In the well-conditioned case, drift falls from 0.400 at $m = 64$ to 0.102 at $m = 512$; for the ill-conditioned encoder it falls from 0.446 to 0.122; for the ill-conditioned head it falls from 0.346 to 0.077. Relative parameter movement shows the same trend, decreasing from roughly 0.15–0.17 at $m = 64$ to about 0.05 at $m = 512$. Thus the wide adapters used to interpret $\lambda_{\min}(K_0)$ are also the settings where the nonlinear model stays closest to its initialization kernel.

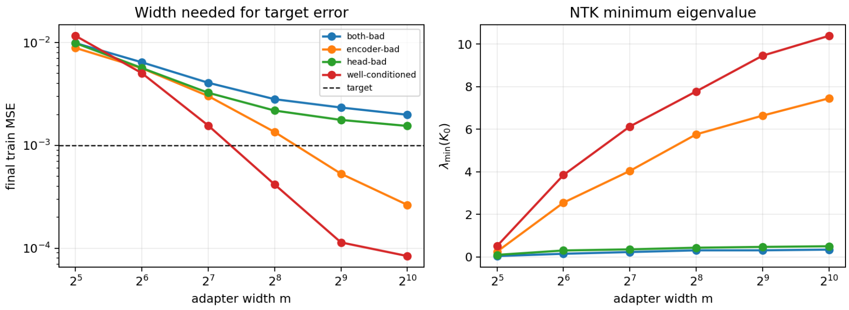


**Figure 3.** E5 width-threshold experiment. The dashed line marks train MSE $10^{-3}$. The required width depends strongly on the frozen-module pair, and conditions with persistently small $\lambda_{\min}(K_0)$ require more width.

# 6 Discussion

The experiments support a specific interpretation of adapter optimization. At very small width, the student is simply underparameterized, so width dominates the error. Once width is moderately large, the adapter has enough random features to fit the teacher map, and the frozen modules become the main source of variation. This explains why $\lambda_{\min}(K_0)$ is most predictive within the wide slices of E3 and E4. In that regime, the model is not failing because it lacks parameters; it is failing because the gradient directions induced by the frozen encoder and head do not span the output residuals equally well. E5 converts this spectral view into a width requirement: better frozen spectra reach the same error budget with a smaller adapter, while poorly conditioned heads remain difficult even after increasing width to 1024.

# 7 Conclusion and Limitations

This project gives a controlled account of adapter optimization between frozen modules. The main result is qualitative but robust: adapter width is necessary, but the conditioning of the frozen representation and frozen head affects the empirical NTK spectrum and the remaining optimization error. The joint-conditioning experiment verifies that the two frozen modules are not independent nuisances; they combine through the same empirical kernel. The width-threshold experiment further shows that this is not only a post-hoc spectral correlation: the width required to hit a fixed error budget changes with the frozen pair. The kernel-drift experiment shows that this diagnostic is most defensible in the moderate-to-wide regime, where the nonlinear adapter remains closer to its initialization kernel. Together, these results explain in a tractable model why different frozen encoder–LLM pairs may require different adapter widths even when the adapter architecture is unchanged.

The limitations are also clear. The experiments use Gaussian representations and fixed linear heads, not real audio encoders or LLMs. The binary success threshold $10^{-4}$ was too strict under the conservative learning-rate budget, so the final analysis relies on continuous training-error metrics. Future work should test the same NTK diagnostics on cached representations from real speech encoders and on nonlinear frozen heads with more realistic local Jacobian spectra.

# References

S. S. Du, J. D. Lee, H. Li, L. Wang, and X. Zhai. Gradient descent finds global minima of deep neural networks. In *International Conference on Machine Learning (ICML)*, 2019.

A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Y. Lei, S. He, J. Hu, D. Zhang, X. Luo, D. Zhu, S. Feng, R. Liu, J. He, Y. Sun, H. Wu, and H. Wang. MoE adapter for large audio language models: Sparsity, disentanglement, and gradient-conflict-free. *arXiv preprint arXiv:2601.02967*, 2026.

S.-w. Yang, P.-H. Chi, Y.-S. Chuang, C.-I. J. Lai, K. Lakhotia, Y. Y. Lin, A. T. Liu, J. Shi, X. Chang, G.-T. Lin, T.-H. Huang, W.-C. Tseng, K.-t. Lee, D.-R. Liu, Z. Huang, S. Dong, S.-W. Li, S. Watanabe, A. Mohamed, and H.-y. Lee. SUPERB: Speech processing universal PERformance benchmark. In *Interspeech*, 2021. arXiv:2105.01051.

J. Zhang, H. Dinkel, Y. Niu, C. Liu, S. Cheng, A. Zhao, and J. Luan. X-ARES: A comprehensive framework for assessing audio encoder performance. In *Interspeech*, 2025. arXiv:2505.16369.